

Test Notes: A Variance-Hierarchy Ablation of the KNS Estimator

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Setup

Kozak, Nagel, and Santosh (2020) estimate the stochastic discount factor from characteristic-managed portfolios with sample mean returns $\bar{\mu}$ and covariance matrix Σ . The L2 estimator is

$$\hat{b} = (\Sigma + \gamma I)^{-1} \bar{\mu}, \quad \gamma = \frac{\text{freq} \cdot \text{tr}(\Sigma)}{T \kappa^2},$$

where κ is selected by contiguous 3-fold cross-validation to maximize out-of-sample cross-sectional R^2 . In principal-component coordinates, where $\Sigma = E \Lambda E'$ and $\Lambda = \text{diag}(d_1, \dots, d_N)$, the coefficient on PC j is

$$\hat{b}_{P,j} = \frac{\bar{\mu}_{P,j}}{d_j + \gamma}.$$

The implied shrinkage factor is $d_j/(d_j + \gamma)$. High-variance PCs are shrunk less, while low-variance PCs are shrunk more. This is the central geometric feature of KNS. The estimator does not merely shrink the SDF. It uses the covariance eigenvalue hierarchy as a prior over which directions deserve more trust in finite samples.

Data and implementation. The test uses the paper’s 50 anomaly-managed portfolio universe from Kozak’s data, daily observations from November 1, 1973 through December 29, 2017. Returns are market-orthogonalized by regressing each portfolio on the CRSP value-weighted excess return with an intercept and taking residuals. The covariance matrix is estimated using Kozak’s Wishart-regularized `regcov.m`,

$$\hat{\Sigma} = a \frac{\text{tr}(\hat{S})}{N} I + (1 - a) \hat{S}, \quad a = \frac{N}{N + T},$$

and the L2 penalty, fold construction, and cross-validation objective follow the released MATLAB code. This matters because the code parameterization of κ includes the $1/T$ normalization above. In this scale the KNS paper reports an optimum near $\kappa \simeq 0.30$.

The empirical covariance matrix is extremely ill-conditioned. In the raw anomaly space, the largest eigenvalue is about 1536 times the smallest. This condition number is not reported as an exact number in the KNS paper, but the paper does recognize the same empirical scale. In Appendix A it notes that the ratio of the largest to smallest eigenvalue is of order 10^3 ,

which implies orders of magnitude more sampling variance for coefficients attached to small-eigenvalue PCs. The 1536 value is the corresponding diagnostic in this replication. It is the finite-sample problem KNS is built to solve. Any estimator close to OLS must invert $\hat{\Sigma}$, so small-eigenvalue directions are amplified by $1/d_j$. The heterogeneous KNS prior counteracts that instability by applying much stronger shrinkage to those directions. Full whitening reduces the transformed spectrum ratio to 1.73, but it also removes the variance hierarchy that tells the ridge estimator where the dangerous inversion directions are.

Motivation

The test asks how much KNS benefits from the covariance eigenvalue hierarchy when pricing a large anomaly cross-section. A flat-spectrum latent representation is attractive for a later, higher-complexity model because it prevents the model from inheriting variance rank as a built-in notion of relevance. But KNS uses variance rank precisely as a finite-sample regularization device. If flattening the spectrum weakens the linear estimator, that does not mean whitening destroys population pricing information. It means the linear estimator loses a useful statistical guide.

This distinction sets up the core interpretation. Whitening preserves the population opportunity set under an invertible linear transformation, but it changes the finite-sample geometry faced by the estimator. The experiment therefore tests the cost of operating without variance-rank guidance. That cost is the baseline obstacle a more complex model must overcome if it is trained on a whitened or partially whitened representation.

Whitening and the Finite-Sample Problem

Let R be the vector of excess returns with mean μ and covariance Σ . The population maximum squared Sharpe ratio is

$$SR_{\max}^2 = \mu' \Sigma^{-1} \mu.$$

For a whitening matrix W satisfying $W \Sigma W' = I$, define $\tilde{R} = WR$ and $\tilde{\mu} = W\mu$. Since $\Sigma^{-1} = W'W$,

$$\tilde{\mu}' \tilde{\mu} = \mu' W' W \mu = \mu' \Sigma^{-1} \mu.$$

Thus whitening leaves the population maximum Sharpe ratio unchanged. It is a coordinate transformation, not a removal of the investment opportunity set.

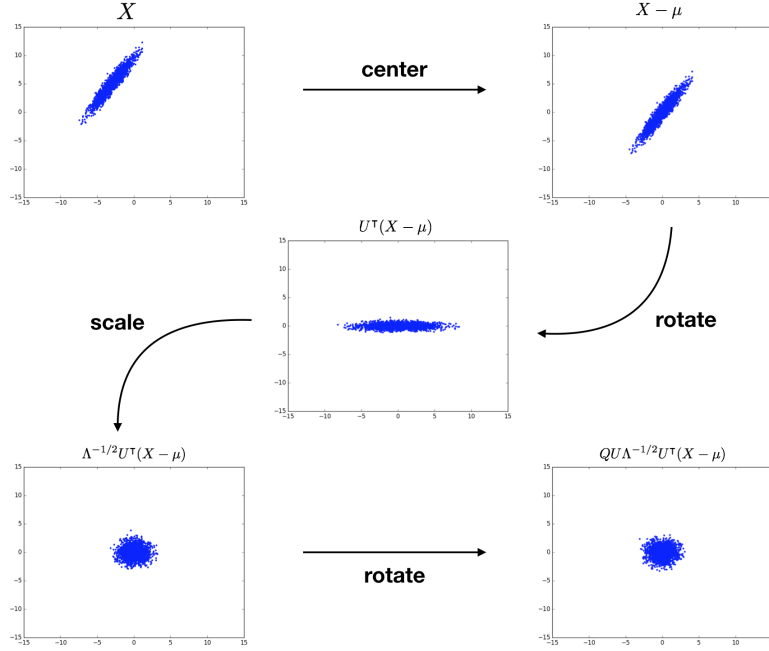


Figure 1: Whitening maps the risk ellipsoid into a sphere. The population opportunity set is preserved, but the finite-sample ridge estimator no longer sees the original eigenvalue hierarchy.

The problem is finite-sample regularization. In PC coordinates, full whitening sets every transformed eigenvalue to one. KNS then applies the same shrinkage to every direction, because the denominator no longer varies with d_j . After mapping the whitened coefficient vector back to the original asset space, the estimator is a rescaled $\hat{\Sigma}^{-1}\bar{\mu}$ portfolio. That is the Pástor–Stambaugh level-shrinkage family. It shrinks all directions uniformly but does not differentially discipline the small-eigenvalue PCs.

The partial-whitening continuum makes this precise. For

$$A(\alpha) = E\Lambda^{-\alpha/2}E', \quad \tilde{R}_\alpha = A(\alpha)R, \quad \alpha \in [0, 1],$$

$\alpha = 0$ is raw KNS and $\alpha = 1$ is full ZCA whitening. The transformed covariance has eigenvalues $d_j^{1-\alpha}$. Increasing α progressively flattens the spectrum and removes the degree to which KNS can use variance rank as a shrinkage guide. All headline R^2 numbers below are evaluated in the original return space after mapping coefficients back, so the comparison is not driven by changing the metric.

Part I: KNS Replication Anchor

The replication anchor matches the paper’s methodology and scale. The data have 11,060 daily observations and 50 anomaly portfolios. Contiguous 3-fold CV selects

$$\kappa^* = 0.3069, \quad \text{CV OOS } R^2 = 26.35\%.$$

The selected κ matches the paper’s stated optimum of roughly 0.30. The top raw factors by absolute t -statistic are industry relative reversals (low volatility), industry momentum-reversals, industry relative reversals, seasonality, and earnings surprises, matching the Table 1a ordering. In PC space, the leading factors are PC4, PC1, PC9, and PC2, the same high-variance block emphasized by KNS Table 1b. This establishes that the ablation is anchored to the KNS result before changing the covariance geometry. It also confirms that the high condition number is not an incidental nuisance. Without the KNS eigenvalue-weighted prior, the same covariance geometry would make the mean-variance inversion unstable.

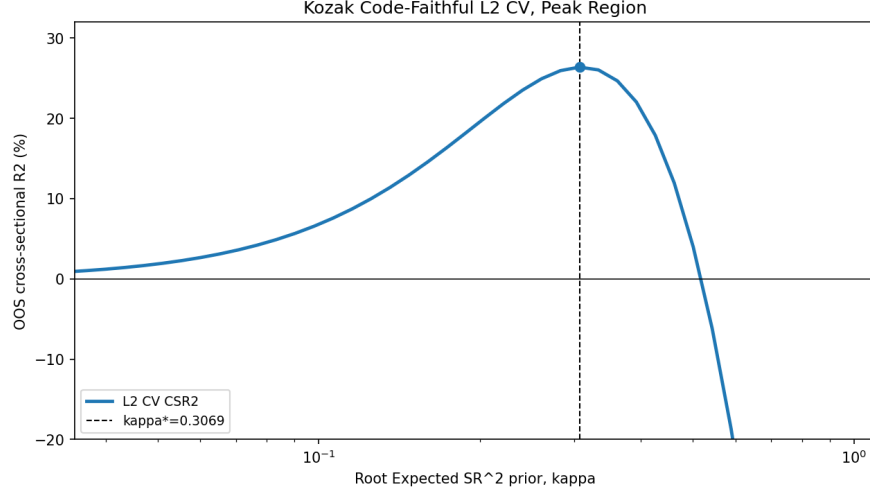


Figure 2: KNS L2 model-selection curve on the paper-50 anomaly universe. The optimum is $\kappa^* = 0.3069$, with peak CV OOS cross-sectional $R^2 = 26.35\%$.

Partial-Whitening Results

The main result is that the raw KNS geometry is essentially optimal for cross-sectional pricing. A very mild amount of whitening is harmless, but substantial whitening steadily degrades original-space R^2 .

α	κ^*	CV R^2	Native R^2	Full-sample SR	OOS SR
0.00	0.3069	26.35%	26.35%	2.236	0.693
0.10	0.3330	26.44%	24.65%	2.428	0.672
0.25	0.3613	26.24%	21.83%	2.692	0.644
0.50	0.4253	24.17%	16.99%	3.133	0.640
0.75	0.5006	18.71%	13.56%	3.505	0.670
1.00	0.5892	10.36%	12.81%	3.692	0.672

The best point on the full grid is $\alpha = 0.10$, with CV $R^2 = 26.44\%$. This is only 0.09 percentage points above raw KNS. By contrast, full whitening falls to 10.36%, a loss of 15.99 percentage points relative to $\alpha = 0$. The conclusion is therefore not that a tiny rescaling of the spectrum is fatal. The conclusion is that the eigenvalue hierarchy is load-bearing. Once the spectrum is materially flattened, KNS loses the differential shrinkage structure that makes it work.

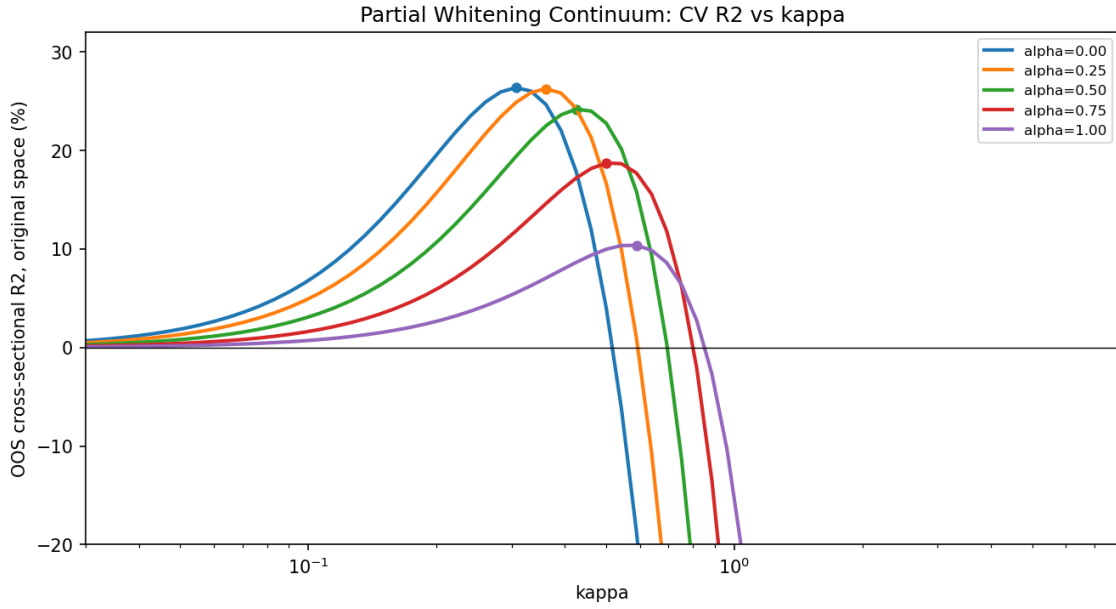


Figure 3: Original-space CV R^2 curves for selected partial-whitening values. The graph is zoomed to the economically relevant region. Extreme OLS-like failures at high κ are intentionally not shown.

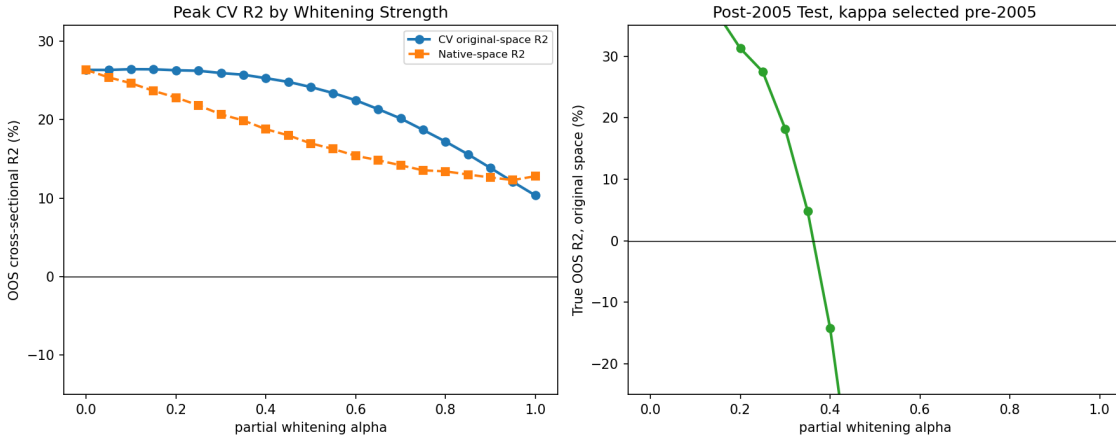


Figure 4: Peak CV R^2 and post-2005 true OOS R^2 by whitening strength. Original-space pricing accuracy is stable for very small α but deteriorates sharply as the covariance spectrum is flattened.



Figure 5: Heatmap of original-space CV R^2 over (α, κ) , clipped to the relevant range. α increases from bottom to top and κ increases from left to right. The white markers show the CV optima for $\alpha = 0$ and $\alpha = 1$, so this panel is separate from the post-2005 true OOS diagnostic.

Sharpe Ratios

The Sharpe-ratio result is deliberately separated from the pricing- R^2 result. Full-sample SR rises monotonically with whitening.

$$2.236 \quad (\alpha = 0) \quad \longrightarrow \quad 3.692 \quad (\alpha = 1).$$

This is not evidence that whitening improves the pricing model. As the estimator moves toward the $\hat{\Sigma}^{-1}\bar{\mu}$ family, it becomes closer to the sample mean-variance portfolio, which is naturally high-SR in sample.

The post-2005 OOS SR is much flatter.

$$0.693 \quad (\alpha = 0), \quad 0.644 \quad (\alpha = 0.25), \quad 0.640 \quad (\alpha = 0.50), \quad 0.672 \quad (\alpha = 1).$$

The aggregate trading opportunity therefore survives much more than the cross-sectional pricing fit. This is exactly what the whitening algebra predicts. the population maximum Sharpe ratio is invariant to invertible transformations, but the finite-sample ridge estimator's ability to price each anomaly portfolio asset-by-asset is not invariant.

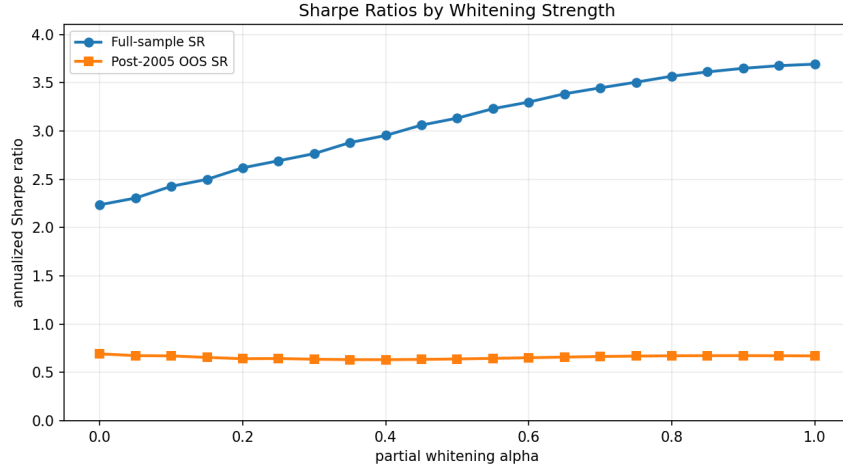


Figure 6: Full-sample and post-2005 OOS Sharpe ratios by whitening strength. Whitening increases full-sample SR, but OOS SR remains comparatively stable even as original-space R^2 collapses.

Interpretation

The partial-whitening continuum sharpens the original binary comparison. The performance loss is not an artifact of a single full-whitening endpoint. As α increases, the spectrum ratio falls from roughly 1536 at $\alpha = 0$ to 1.73 at $\alpha = 1$, and original-space CV R^2 falls from 26.35% to 10.36%. This is the central empirical scale of the exercise. The original covariance matrix is so ill-conditioned that KNS is doing substantial stabilization work. flattening the spectrum removes both the conditioning problem and the information KNS uses to decide where stabilization is needed. The first tenth of whitening is effectively neutral, but the cost becomes clear once the variance hierarchy is meaningfully compressed.

The post-2005 diagnostic tells the same story in a more severe form. With κ selected only on pre-2005 data, raw KNS obtains 44.02% true OOS original-space R^2 . Partial whitening at $\alpha = 0.25$ remains positive at 27.49%, but $\alpha = 0.50$ falls to -77.84% , and full whitening is deeply negative. These extreme negative values should not be read as a literal claim that full whitening is economically impossible. They show that a finite-sample linear ridge estimator becomes unstable as the covariance hierarchy is removed and the back-transformed coefficient vector inherits the low-variance amplification of $\hat{\Sigma}^{-1}$.

The SR evidence is the counterweight. OOS SR remains in a narrow band even when pricing R^2 deteriorates. A flat-spectrum representation can still contain economically meaningful aggregate return opportunities. What it loses is the linear estimator's ability to use variance rank as a disciplined prior for cross-sectional pricing. That is the point of the ablation. The low-dimensional linear KNS estimator needs the eigenvalue hierarchy, but the persistence of Sharpe suggests that a richer model may still recover useful structure from a variance-blind representation.

Takeaways

The experiment establishes three facts.

First, the Part I replication is anchored to KNS. The clean implementation selects $\kappa^* = 0.3069$ and reaches 26.35% CV OOS R^2 on the paper-50 daily anomaly universe, matching the scale and factor ordering in the paper.

Second, the variance hierarchy is load-bearing for the linear ridge estimator. The raw covariance condition number is roughly 1536, so OLS-style inversion is catastrophically unstable without differential shrinkage. Full whitening reduces the transformed spectrum ratio to 1.73, but original-space CV R^2 falls from 26.35% to 10.36%. The partial-whitening continuum shows that performance degrades as the covariance spectrum is flattened and the KNS stabilizing prior is removed.

Third, Sharpe ratio and cross-sectional pricing accuracy separate. Full-sample SR rises with whitening and post-2005 OOS SR is relatively stable, but original-space R^2 collapses. For the complex model down the line, this is the opening. A variance-blind representation is a harder statistical environment than KNS, but it may preserve aggregate economic signal that a more expressive pricing head can learn to organize.